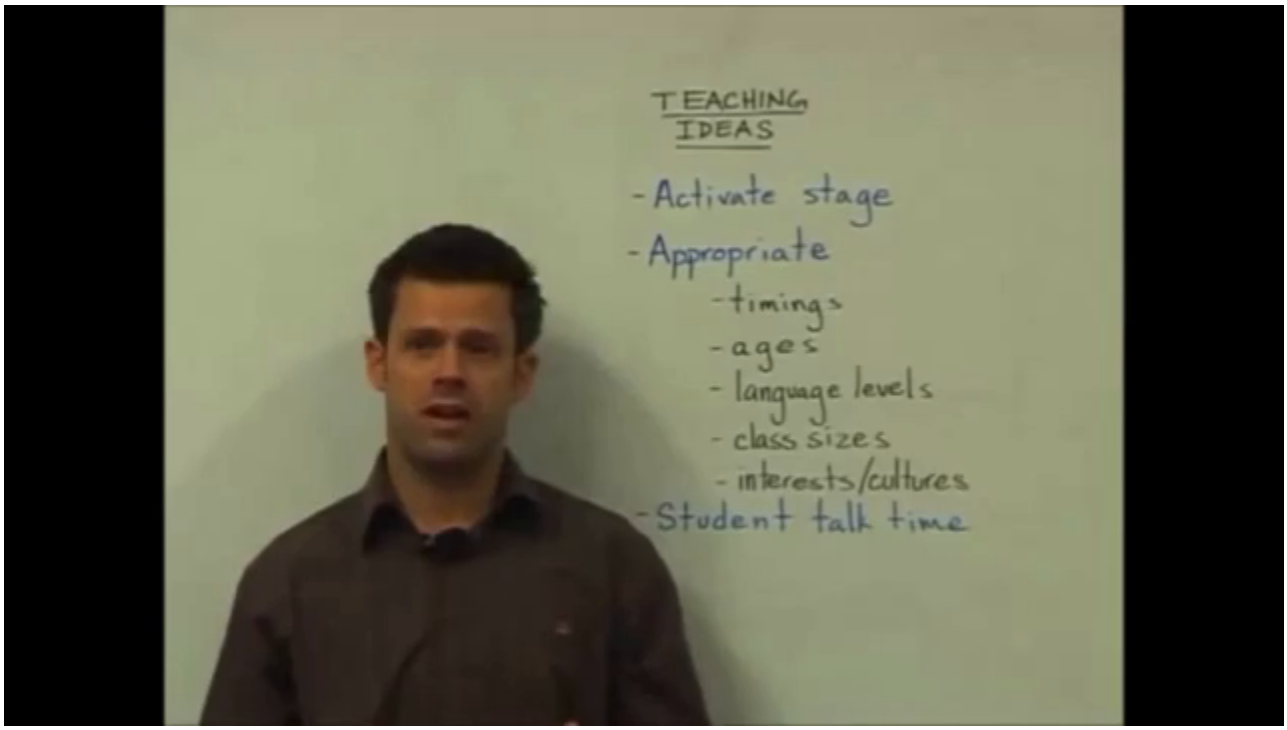
Frame Review: Keep the Lesson

Joseph Ayanda / AI-assisted development / 23 October 2026

Local qualification complete. Revised AWS deployment is deferred until the account owner is home. No natural lesson repair or teacher benefit is claimed.



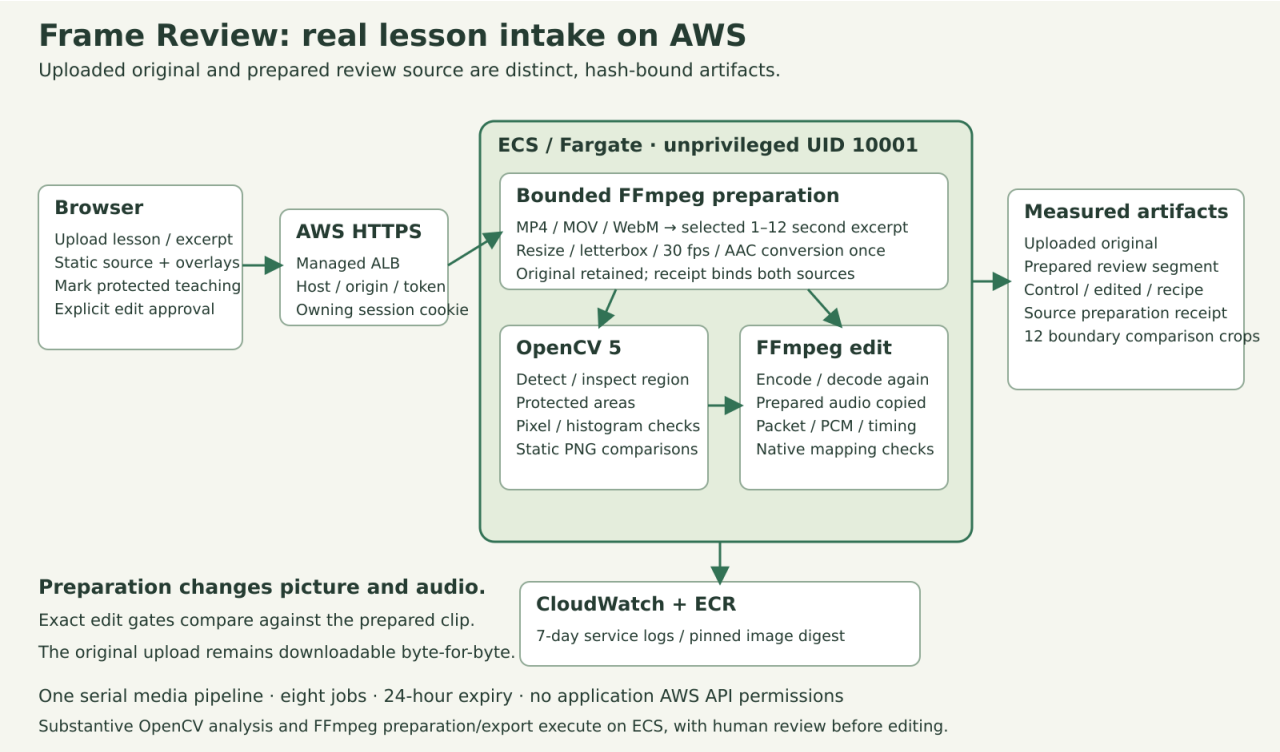
Real ITTT teacher footage, human narration. TEFL & TESOL Courses - ITTT, Present Simple - Teaching Ideas - 1 (2013), CC BY 3.0, via Wikimedia Commons. Selected excerpt, resize and format conversion. No endorsement.

Problem and user. A teacher or lesson editor has a flattened recording after losing its editable project. A repeating decoration competes with the explanation. Slowing the whole video also changes the teacher, gestures and narration. The useful decision is smaller: which visual detail may change, and what teaching must stay on its original timeline?

Working interaction. Upload a supported ordinary lesson, select 1-12 seconds, check the prepared teaching detail, save a teaching area and inspect the proposed local 5x retime. Explicit approval binds source, prediction and recipe. The actual encoded result must pass its preservation checks. Keeping a clip unchanged or refusing an unsupported edit is an explicit, downloadable decision.

Measured positive. A disclosed added-corner challenge on real classroom footage passed all 360 teacher/whiteboard RGB comparisons, native outside/mapping checks and prepared narration identity. Its added defect is controlled. Naturally occurring lesson repair remains unqualified.

Vision drives a bounded decision



Substantive OpenCV 5. OpenCV measures luminance changes and connected candidate geometry, generates overlays and encoded-frame crops, compares RGB pixels and accumulates error histograms. NumPy represents native YUV planes and frame mappings. FFmpeg performs bounded demuxing, decoding, preparation, encoding and audio checks. No trained semantic classifier or autonomous agent is claimed.

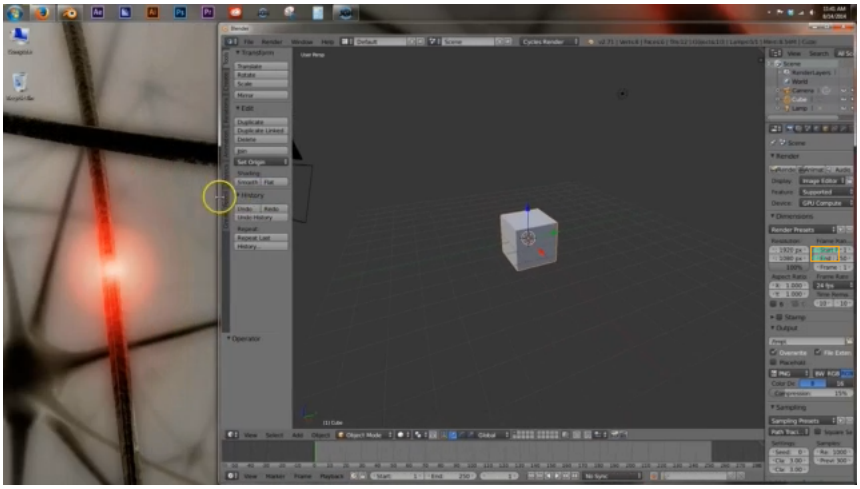
Preservation contract. A prepared lesson must declare at least one fixed teaching area; the server refuses an empty plan. Protection overlapping an edit is refused. Exact local-retime checks compare every protected RGB frame with the prepared input; outside native samples and the selected native mapping are independently checked. Prepared AAC packets, timestamps, skip metadata and decoded samples must remain unchanged. Failed checks withhold the edited download.

Preparation changes media. The selected excerpt is resized/letterboxed to 640x360, converted to 30 fps and encoded once to H.264 and optional AAC-LC mono/48 kHz. This can make small labels unreadable. A full-size static prepared frame and explicit readability check make that limitation visible. Exact edit checks do not establish original-video fidelity or readability. Original upload and prepared source have separate hashes and downloads.

Limits of the visual rule. At least seven alternating luminance transitions in a rolling second, connected support at least 64 pixels. This can miss ordinary blinking or notifications and can flag useful changing settings. A candidate is a region to review, not a semantic distraction label. Repetition is separately gated; no threshold was relaxed to turn natural failures into successes.

Natural recordings changed the product

Evidence group	Observed outcome and interpretation
Four new original tutorials	Wikilink, Tamil software installation, Blender and interface tutorials. Identical 12-second windows at 12, 40 and 68 seconds, selected before testing. Twelve real HTTP workflows passed; eleven returned no candidate; one flagged Blender settings. All were left unchanged.
Original classroom + citations	Two further original excerpts at 40-52 seconds. No candidate and no edit. Together: six original lesson sources, fourteen selected unaugmented excerpts. Not a representative benchmark.
Blender settings candidate	[605,181,22,12], 114 pixels, frames 40-315. Protecting the settings column refused the plan before approval. Useful teaching content must not be relabeled unwanted simply because it changes.
Natural camera-flicker stress	Separate lecture-hall lamp recording, not an active lesson. Commons 1080p transcode; uploader original is too large. Two fixed excerpts produced 3 and 5 candidates. Largest-support attempts both failed the unchanged recurrence gate.
Controlled real-teacher positive	A deliberately added 32x32 flashing corner. Explicit plan and approval; accepted measured export. This establishes functional preservation, not natural-defect repair success.



A detected field in the Blender lesson. Lorenzoh, Module 3 Lesson 1 Screencast v2, CC BY-SA 4.0. Changes: excerpt, resize, encode and candidate overlay. Derivative image distributed under CC BY-SA 4.0. Full source/license URLs in the natural audit.

Failure recovery fixed. A refused prepared export previously hid its source links. It now offers hash-verified uploaded/prepared files, the preparation receipt and a failure-aware lesson decision record; the rejected edit remains unavailable. Both natural camera failures passed these recovery checks. No-candidate records also explain what the detector cannot establish.

Qualification and competition delivery

Final local image. All ten media stages passed from the start: 137 backend unit tests, numerical/duration/refusal suites, audio/silent exports, authored lesson, real-classroom intake/controlled challenge and 65 studio checks. Linux studio checks pass with one platform skip; Windows passes the same 65 checks with its documented platform skip. An earlier new test incorrectly expected HTTP 400 instead of the correct 409 state conflict; the failed run is retained and the expectation was corrected before the final full rerun.

Additional actual checks. The fixed twelve-excerpt cohort was repeated on this image with identical prepared hashes and detected regions. The independent two-source/classroom challenge passed. Actual local browser checks cover the original Blender refusal, source/decision downloads, controlled classroom approval, all twelve loaded PNG crops, report/output/independent hash agreement, reload recovery, 375/890 layouts and natural-camera refusal recovery. Owning-session HTTP crop hashes match; unowned job/source/decision/edited paths return 404. Playback observations are recorded separately; none of these are a teacher trial.

AWS delivery. The earlier authored-lesson workflow remains on ECS/Fargate behind managed HTTPS, with ECR images and seven-day CloudWatch logs. The revised image has not replaced it. Its locally tested adapter runs preparation, OpenCV analysis and FFmpeg export in one serial pipeline, as UID 10001 without an application AWS API role. Cloud-adapter labels in local screenshots do not prove AWS execution. The actual new HTTPS workflow must be checked after deployment. The candidate archive includes this source and report. Its actual Devpost attachment identity is recorded separately from runtime qualification.

Admission and operation. MP4/MOV or WebM up to 64 MiB; one supported 8-bit SDR video up to 3840x2160 and at most one supported mono/stereo audio track; sources up to 30 minutes. Excerpt starts within 600 seconds and lasts 1-12 seconds. No full-lesson splice-back. File/pipe protocols and process, wall-time, memory, disk and job caps bound processing. Owner cookies and host/origin/request-token checks scope access. Task replacement can clear jobs. Known package findings remain; restrictions are mitigations, not certification.

Impact and next evidence. No qualified natural lesson repair, teacher time saving, learning outcome or adoption result exists yet. Test held-out defective lessons with creators against competent manual editing, measuring successful tasks, review burden and unintended changes. The competition weights technical execution 30% and impact 20%; preserving negative evidence supports the former and cannot substitute for the latter. The general prize path is used; no COOL or Agentic Vision award claim. Joseph will use his real voice for a later final demo; the earlier 2:46 video uses disclosed neural narration.

Exact runtime files: 87
Source manifest: 54468669f8704929c1085a3bf2276cc6dde0ad31390b80de065c35e4c5c26aa6
Local Docker image ID: sha256:050ea28fdbd0d2dc8afe474a6a725a8978a51086f144784ad3e62fd672783c0d
Code, deployment instructions, all receipts and source/license records: https://github.com/josepha-mayo/opencv-frame-review
Official requirements: https://opencv26.devpost.com/